

NONCLINICAL REPORT

Nonclinical Report No. VRC07

Evaluation of Immunogenicity and Efficacy from an Expanded Dose Range of mRNA-1273 in
Rhesus Macaques

SPONSOR:

Vaccine Research Center
National Institute of Allergy and Infectious Diseases
National Institutes of Health
Bethesda, MD 20892

PERFORMING LABORATORY:

Viral Pathogenesis Laboratory
Vaccine Research Center
National Institutes of Health
Bethesda, MD 20892

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03 November 2020

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LIST OF ABBREVIATIONS

Abbreviation	Definition
ACE-2	angiotensin-converting enzyme 2
ATCC	American Type Culture Collection
AUC	area under the curve
BAL	bronchoalveolar lavage
CoV	coronavirus
COVID-19	coronavirus disease 2019
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine
ECL	electrochemiluminescence
ELISA	enzyme-linked immunosorbent assay
HEK	human embryonic kidney
ID ₅₀	half-maximal inhibitory dilution
Ig	immunoglobulin
IL	interleukin
IM	intramuscular
IN	intranasal(ly)
IT	intratracheal(ly)
LNP	lipid nanoparticle
mRNA	messenger RNA
MIMOSA	Mixture Models for Single-Cell Assays
(b) (4)	
NIAID	National Institute of Allergy and Infectious Disease
NIH	National Institutes of Health
NS	nasal swab(s)
NTD	N-terminal domain
NTFIX	noncoding mRNA
PBS	phosphate-buffered saline
PCR	polymerase chain reaction
PEG	polyethylene glycol
PEG2000-DMG	1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol
PFU	plaque-forming unit(s)
RBD	receptor binding domain
RT-qPCR	reverse transcription quantitative PCR
S	spike
S1 NTD	N-terminal domain of S1
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
SARS-CoV-2	2019 novel coronavirus
sgRNA	subgenomic RNA
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate
Tfh	T-follicular helper

Abbreviation	Definition
Th1	T-helper 1
Th2	T-helper 2
Tris HCl	tris(hydroxymethyl)aminomethane hydrochloride
USAMRIID	United States Army Medical Research Institute of Infectious Disease
UTR	untranslated region
VRC	Vaccine Research Center
WT	wild type

QUALITY ASSURANCE STATEMENT

The study was not conducted according to Good Laboratory Practice regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized S protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic.

The objective of Study VRC07 was to evaluate the protective efficacy of mRNA-1273 against wild type (WT) SARS-CoV-2 isolate USA-WA1/2020 challenge using rhesus macaques.

ModernaTX, Inc. is developing mRNA-1273, a novel lipid nanoparticle (LNP)-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes for the SARS-CoV-2 S glycoprotein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P). The mRNA-1273 Drug Product consists of the mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to ModernaTX, Inc.'s mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol (PEG2000-DMG), respectively. The test article in this study consisted of mRNA-1273 Drug Product (WH2020_NatSP_2P, Tech dev DP). A noncoding mRNA (NTFIX) containing a short mRNA sequence formulated into the same LNP dispersion as mRNA-1273 and phosphate-buffered saline were used as negative controls.

Rhesus macaques (n = 6/group) were vaccinated with either 2.5 or 30 µg of mRNA-1273, NTFIX, or PBS via intramuscular (IM) injection on a prime (Week 0)/boost (Week 4) schedule. Additionally, 100 µg of mRNA-1273 was given at Week 0 without a boost to assess prime-only efficacy. At Week 8 (4 weeks post-boost), the rhesus macaques were challenged intratracheally (IT) and intranasally (IN) with a total dose of 7.6×10^5 plaque-forming units (PFU) of SARS-CoV-2 isolate USAWA1/2020. Pre-challenge serum samples were collected at Week -1, Week 0, Week 2, Week 4, Week 6, and Week 7 for serology and T-cell analysis. Post-challenge serum samples were collected on Day 2, Day 4, and Day 7; nasal swab (NS) samples were collected on Day 1, Day 2, Day 4, and Day 7; and bronchoalveolar lavage (BAL) samples were collected on Day 2, Day 4, and Day 7. Lung sections were collected for histopathological analysis and viral quantification on Day 7, Day 8, or Day 9 post-challenge.

Antibody responses were assessed pre- and post-challenge, and protection after challenge was assessed in the upper and lower airways from NS samples and BAL fluid by polymerase chain reaction (PCR) to measure viral RNA. Reverse-transcription quantitative PCR (RT-qPCR) using

primers specific for subgenomic RNA (sgRNA) were used to quantify nascent viral replication. Assessment of histopathology and viral quantification were performed on lung tissue specimens.

In rhesus macaques, mRNA-1273 elicited robust immunoglobulin (Ig) G binding antibody responses in a dose-dependent manner, with post-boost antibody titers that were higher than post-prime titers in the 2.5 µg and 30 µg groups. All mRNA-1273-vaccinated treatment groups had S-specific binding antibody and angiotensin converting enzyme-2 (ACE-2) competing antibody levels that were similar to or higher than those of human convalescent sera. Administration of mRNA-1273 elicited higher antibody responses in the receptor binding domain (RBD) and the N-terminal domain (NTD) of S1 (S1_NTD) in the 30 µg prime/boost and 100 µg prime-only groups than in the 2.5 µg prime/boost group.

A robust neutralizing antibody response against SARS-CoV-2 pseudovirus was observed post-boost in all mRNA-1273 groups and was observed post-prime in the 30 µg prime/boost group. Neutralizing activity measured against SARS-CoV-2 pseudovirus and live virus was evident in the 30 µg prime/boost and 100 µg prime/boost groups, both of which also demonstrated the strongest antibody response compared with human convalescent sera (data from the 100 µg prime/boost group was provided from [Report VRC04](#)). In addition, neutralizing activity assessed post-challenge showed that neutralizing antibody titers were substantially higher in the 30 µg mRNA-1273 prime/boost group at 2, 4, and 7 days post-challenge than in the control group and the 2.5 µg prime/boost group and were somewhat higher than the 100 µg prime-only group at 7 days post-challenge.

Activated CD4⁺ T cells can be segregated into T-helper 1 (Th1)- and T-helper 2 (Th2)-directed responses based on the production of specific cytokines and are critical for B cell activation and antibody production. Importantly, the induction of CD4⁺ Th2 T-cell responses (production of interleukin [IL]-4, IL-5, or IL-13) has been linked to vaccine-associated enhanced respiratory disease (ERD), as seen in some human respiratory virus infections (eg, respiratory syncytial virus [[Kim et al 1969](#)], measles [[Polack 2007](#)], and animal models of SARS-CoV infection [[Bolles et al 2011](#); [Corbett et al 2020](#); [Deming et al 2007](#)]); however, vaccine-associated ERD has not been observed with exclusive CD4⁺ Th1 T-cell responses (production of interferon-γ, IL-2, and tumor necrosis factor-α) ([Grifoni et al 2020](#); [Peng et al 2020](#); [Sekine et al 2020](#); [Weiskopf et al 2020](#)). Vaccination with mRNA-1273 induced Th1-directed CD4⁺ T-cell and interleukin (IL)-21-producing T-follicular helper (Tfh) cell responses with no evidence of Th2-directed CD4⁺ T-cell responses.

Post-challenge, quantification of SARS-CoV-2 viral load using PCR demonstrated that all mRNA-1273-immunized groups had lower viral loads in BAL fluid than control animals. There were no substantial differences in viral loads in the NS samples of mRNA-1273-vaccinated and control animals, although animals in the 30 µg prime/boost group trended towards having lower

viral titers than those in the other groups. Measurement of sgRNA levels after challenge showed that mRNA-1273 was protective in the lungs and nose of animals in the 30 µg prime/boost and 100 µg prime-only group: no animals in the 100 µg prime-only group and 1 of 6 animals in the 30 µg prime/boost group had detectable sgRNA in BAL fluid after challenge. In NS samples, no animals in the 30 µg prime/boost or the 100 µg prime-only group had detectable sgRNA at 1 and 4 days post-challenge and only 1 of 6 animals in the 30 µg prime/boost group and 2 of 6 animals in the 100 µg prime-only group had detectable levels of sgRNA at 2 days post-challenge. Limited to no lung inflammation was measured post-challenge in animals in the mRNA-1273 groups, whereas moderate to severe inflammation was observed in the lungs of control animals. No evidence of vaccine-associated immunopathology was found in histopathological analysis of lung tissue and no viral antigen was detected in the 30 µg prime/boost group post-challenge.

Results from this study showed that mRNA-1273 elicits robust binding and neutralizing antibody responses in rhesus macaques, especially in the 30 µg prime/boost group. Similar antibody responses were observed in the RBD and NTD domains of the S1 subunit. Vaccination with mRNA-1273 induced Th1-directed T-cell and IL-21-producing Tfh cell responses, a profile that is not predicted to promote ERD. mRNA-1273 protected in the lungs and nose of animals in the 30 µg prime/boost and 100 µg prime-only groups, as measured by sgRNA. No evidence of vaccine-associated immunopathology was found in histopathological analysis of lung tissue of animals vaccinated with mRNA-1273, and no viral antigen was detected in the 30 µg prime/boost group post-challenge.

Overall, immunization of rhesus macaques with mRNA-1273 induced robust SARS-CoV-2 binding antibodies and neutralizing activity and protection in the upper and lower airways from WT SARS-CoV-2 challenge, with no pathologic changes observed in the lungs or signs of disease enhancement.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

Vaccine Research Center (VRC)
National Institute of Allergy and Infectious Diseases (NIAID)
National Institutes of Health (NIH)
Bethesda, MD 20892

2.1.2. Performing Laboratory

Viral Pathogenesis Laboratory, VRC, NIH
Building 40 Room 2608
Bethesda, MD 20892

Vaccine Immunology Program
9 West Watkins Mill Road
Gaithersburg, MD 20878

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2.1.3. Study Number

Study number VRC-20-857.2 (VRC07).

2.1.4. Purpose of the Study

The objective of this study was to evaluate the protective efficacy of mRNA-1273 against WT SARS-CoV-2 isolate USA-WA1/2020 challenge using rhesus macaques.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved by the Animal Care and Use Committee of the VRC (Protocol VRC-20-857) and conducted in compliance with US NIH regulations.

2.1.6. Study Director**(b) (6)**

Viral Pathogenesis Laboratory, VRC, NIH

Building 40 Room 2608

Bethesda, MD 20892

2.1.7. Key Personnel

Role	Study VRC07
Study coordinator	(b) (4), (b) (6)
Analytical assays: ELISA	(b) (6) Vaccine Research Center, NIAID, NIH Bethesda, MD 20892
Analytical assays: ACE-2 binding inhibition	McDermott Lab Vaccine Immunology Program 9 West Watkins Mill Road Gaithersburg, MD 20878
Analytical assays: Pseudovirus neutralization assay	(b) (6) Vaccine Research Center, NIAID, NIH Bethesda, MD 20892
Analytical assays: NanoLuc assay	(b) (4), (b) (6)
Veterinary pathologist	(b) (6) National Institutes of Health Veterinary Pathologist Chief, Infectious Disease Pathogenesis Section 12735 Twinbrook Parkway, Room 2W-01 Rockville, MD 20852-1770
Report compilation	PPD 3900 Paramount Parkway Morrisville, NC 27560

2.1.8. Dates of Study Performance

Schedule of Events	Study VRC07
Dosing Prime Boost	Initial (Week 0): 19 May 2020 Boost (Week 4): 17 June 2020
Challenge with WT SARS-CoV-2	Week 8: 13 July 2020
Pre-challenge samples collected	Serum (Week -1, Week 0, Week 2, Week 4, Week 6, and Week 7)
Post-challenge samples collected	Serum (Day 2, Day 4, and Day 7) Nasal swabs (Day 1, Day 2, Day 4, and Day 7) Bronchoalveolar lavage (Day 2, Day 4, and Day 7) Lung sections (Day 7, Day 8, or Day 9)

2.1.9. Records Maintained

Data are archived in the Viral Pathogenesis Laboratory (VRC, NIH) database, which is password protected and can be accessed only by authorized personnel. The server is maintained by the NIH information technology staff and is backed up remotely on a daily basis

2.2. Test Articles

The test article in this study was 2.5, 30, and 100 µg mRNA-1273. A noncoding mRNA (NTFIX) and PBS were used as negative controls.

2.2.1. mRNA-1273 Test Article

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102, cholesterol, DSPC, and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in this study was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP) prepared with the same method as the Good Manufacturing Practice mRNA-1273 clinical Drug Product ([Appendix 1](#)). This product is a novel LNP-encapsulated, mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes for the full-length S protein of SARS-CoV-2, modified to introduce 2 proline residues to stabilize the S protein into a prefusion conformation (S-2P). mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process described previously ([Hassett et al 2019](#)).

Preclinical mRNA-1273 vaccine was manufactured by ModernaTX, Inc. A sequence-optimized mRNA encoding prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using

an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N¹m-pseudouridine (Nelson et al 2020). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O-methyltransferase (New England Biolabs). The mRNA was purified (b) (4)

(b) (4), buffer exchanged (b) (4)

sterile filtered, and kept frozen at -20°C until further use.

mRNA-1273 was encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as previously described (Hassett et al 2019). (b) (4)

(b) (4)

(b) (4) The mixture was neutralized with tris(hydroxymethyl)aminomethane hydrochloride (Tris HCl), pH 7.5, sucrose was added as a cryoprotectant, and the final solution was sterile filtered. Vials were filled with mRNA-1273-formulated LNP and stored frozen at -70°C until further use. The Drug Product underwent analytical characterization, which included the determination of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin, and bioburden, and the material was deemed acceptable for the in vivo study (Corbett et al 2020).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No	mRNA Lot No.	Size (nm)	PDI	EE%
mRNA-1273 Drug Product	mRNA SARS-CoV-2 spike protein ^a	(b) (4)				

Abbreviations: EE% = encapsulation efficiency; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; S = spike; SARS-CoV-2 = 2019 novel coronavirus; SM-102 = heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate.

Note: All mRNA test material stocks were formulated in SM-102-containing LNPs with a final storage buffer in 10.7 mM sodium acetate; 8.7% sucrose; 20 mM tris(hydroxymethyl)aminomethane hydrochloride at pH 7.5 and stored frozen at -70°C.

The EE% is the percentage of mRNA (b) (4)

(b) (4)

^a Full-length SARS-CoV-2 S protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) that includes a transmembrane domain for insertion into the cell membrane after translation.

^b Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. Control Messenger RNA

The NTFIX contains a short mRNA sequence formulated into the same LNP dispersion as mRNA-1273. After delivery into cells, this mRNA is not translated into protein.

2.2.3. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4) was purchased commercially. One group of rhesus macaques was treated with PBS, but the data are not included in this report because the control mRNA data provide a more relevant comparison than the PBS data and results from PBS and control mRNA were consistent.

2.3. Test System

Experiments were conducted in 30 Indian-origin rhesus macaques (10 female and 20 male; 3 to 6 years old) provided by the VRC and transferred to (b) (4). After SARS-CoV-2 infection, rhesus macaques have transient viral replication in the upper and lower airways and mild to moderate inflammation in the lung that resolves within 14 days (Yu et al 2020). Thus, rhesus macaques are a useful animal model for assessing vaccine-mediated protection against early viral replication.

Human convalescent serum was obtained from adults with mild, moderate, or severe CoV disease 2019 (COVID-19) and a history of laboratory-confirmed SARS-CoV-2 infection. Details are provided in Section 2.4.3.

2.4. Methods

2.4.1. Study Design and Treatment Groups

Rhesus macaques were treated according to the study design presented in Table 2. Rhesus macaques (n = 6/group) were administered 2.5 or 30 µg of mRNA-1273; PBS (data not shown); or 100 µg of NTFIX mRNA on a prime (Week 0)/boost (Week 4) schedule. An additional group of rhesus macaques was administered 100 µg of mRNA-1273 on a prime-only schedule. Animals were vaccinated via IM injection in the right hind leg.

Data from a single cohort of animals from Report VRC04 were included in Figure 2 for comparison. That group was administered 100 µg of mRNA-1273 on a prime (Week 0)/boost (Week 4) schedule.

Sera collected pre-immunization (Week –1 and Week 0; Figure 1 only), post-prime (Week 2 and Week 4), and post-boost (Week 6 and Week 7) were assessed for SARS-CoV-2 S-2P antibody binding, as measured by enzyme-linked immunosorbent assay (ELISA), and neutralizing antibodies, as measured by pseudotyped lentivirus. Sera collected at Week 7 (3 weeks post-boost) were assessed for S-, RBD- and S1_NTD-specific IgG antibody binding, as measured by

ELISA and neutralizing antibodies as measured by live-virus neutralization assay. Inhibition of ACE-2 binding to RBD, as measured by an electrochemiluminescence (ECL) ACE-2 binding assay, was also assessed from sera collected at Week 7 (3 weeks post-boost). Additionally, sera collected on Day 2, Day 4, and Day 7 post-challenge was tested for neutralizing antibodies as measured by pseudotyped lentivirus.

At Week 8 (4 weeks post-boost), animals immunized with mRNA-1273 were challenged with a total dose of 7.6×10^5 PFU from a stock of 1.9×10^5 PFU/mL of SARS-CoV-2 (USAWA1/2020). The challenge was administered in a total volume of 4 mL (3 mL IT and 1 mL IN [0.5 mL per nostril]) SARS-CoV-2. Antibody responses were assessed post-challenge. Bronchoalveolar lavage fluid was collected on Day 2, Day 4, and Day 7 post-challenge and NS samples were obtained on Day 1, Day 2, Day 4, and Day 7 post-challenge. Active viral replication and viral genomes were assessed by PCR and RT-qPCR. Histopathological analysis and viral quantification were performed on lung tissue specimens collected after animals were euthanized at Day 7, Day 8, or Day 9 post-challenge.

Table 2: Treatment Regimen for Study VRC07

Group (n=6)	Treatment	Vaccination Schedule		Immunization Phase Endpoints	Challenge Week 8 ^a	Post-challenge Endpoints
		Prime (Week 0)	Boost (Week 4)			
1	PBS	NA	NA	Binding titers by ELISA Neutralizing activity by pseudovirus and live-virus neutralization assays Th1 vs Th2 T-cell responses by ICS ACE-2 binding inhibition	WT SARS-CoV-2	Clinical evaluation Viral load by PCR and sgRNA by RT-qPCR (BAL fluid and NS) Lung histopathology
2	NTFIX mRNA	100 µg	100 µg			
3	mRNA-1273	2.5 µg	2.5 µg			
4	mRNA-1273	30 µg	30 µg			
5	mRNA-1273	100 µg	NA			

Abbreviations: ACE-2 = angiotensin-converting enzyme-2; BAL = bronchoalveolar lavage;

ELISA = enzyme-linked immunosorbent assay, ICS = intracellular cytokine staining; mRNA = messenger RNA; NA = not applicable, NS = nasal swab; NTFIX = noncoding mRNA; PBS = phosphate-buffered saline; PCR = polymerase chain reaction; RT-qPCR = reverse transcription quantitative polymerase chain reaction; PFU = plaque-forming units; SARS-CoV-2 = 2019 novel coronavirus; sgRNA = subgenomic RNA; Th1 = T-helper 1; Th2 = T-helper 2; WT = wild type.

^a Animals were challenged with a total dose of 7.6×10^5 PFU from a stock of 1.9×10^5 PFU/mL of SARS-CoV-2 (USAWA1/2020). The challenge was administered in a total volume of 4 mL (3 mL intratracheal and 1 mL intranasal [0.5 mL per nostril]) of SARS-CoV-2.

2.4.2. Antibody Measurements

2.4.2.1. Pseudovirus Neutralization Assay

(b) (4)

(b) (4) and neutralization was evaluated as previously described (Pallesen et al 2017). Briefly, triplicate heat-inactivated sera was mixed with pseudoviruses, incubated, and then added to ACE-2-expressing HEK 293T/17 cells. Seventy-two hours later, cells were lysed and luciferase activity (relative light units) was measured. Percent neutralization was normalized considering uninfected cells as 100% neutralization and cells infected with only pseudovirus as 0% neutralization. Reciprocal 50% inhibitory dilution (ID₅₀) titers were determined using a log (inhibitor) versus normalized response (variable slope) nonlinear function in Prism v8 (GraphPad).

2.4.2.2. Serum Antibody Measurements by Enzyme-Linked Immunosorbent Assay

SARS-CoV-2 S-2P, RBD, and S1_NTD-specific IgG antibody titers in sera were determined by ELISA, similar to previously published methods (Corbett et al 2020). Briefly, (b) (4)

(b) (4)) (4)) were coated with SARS-CoV-2 S-2P (100 ng/well) in 1× PBS at 4°C for 16 hours. After standard washes and blocks, plates were incubated with duplicate serial dilutions (1:100, 4-fold, 8×) of heat-inactivated sera for 1 hour at room temperature. Following washes, anti-monkey IgG (H+L) secondary antibody conjugated in horseradish peroxidase in a blocking buffer at a 1:10,000 dilution (b) (4)) was added. 3,5,3'-Tetramethylbenzidine (b) (4)) substrate was used to detect antibody responses. The optical density of each well was read at 450 nm and 650 nm by (b) (4)). Area under the curve (AUC) was calculated using Prism v8 (GraphPad), setting the baseline with a defined endpoint (average of negative control wells + 10STDEV) and taking the total peak area. Area under the curve indicates the amount of IgG binding to S-2P over time. A multi-array ELISA was used to detect RBD and S1_NTD-specific IgG antibodies in sera.

2.4.2.3. ACE-2 Binding Inhibition Assay

(b) (4) 384-well 4-Spot Custom Serology (b) (4) plates precoated with RBD were blocked for 30 minutes with (b) (4) Blocker A (5% bovine serum albumin) at room temperature. The blocking solution was washed off (3 cycles of PBS-Tween), and test samples were applied in duplicate to the wells at 1:40 dilution and allowed to incubate with shaking (1300 rpm) for 1 hour at room temperature. (b) (4) Assay Diluent 100 in

quadruplicate was used as the negative control. (b) (4) labeled ACE-2 (supplied at 200×) was applied to the wells and allowed to associate with the test samples and RBD within the assay wells for 1 hour at room temperature with shaking (1300 rpm). Plates were then washed (3 cycles of PBS-Tween) to remove unbound detection antibody. (b) (4) (b) (4) read buffer substrate was applied to the wells, and the plate was entered into the (b) (4) detection system.

A current was applied to the plate and areas of well surface where RBD had complexed with (b) (4) -ACE-2 emitted light in the presence of the ECL substrate. The (b) (4) (b) (4) detection system quantitated the amount of light emitted and reported the ECL unit response as a result for each test sample and control of the plate. The amount of signal emitted in wells containing no sample (assay diluent only) was evaluated as the maximal binding response. Reduction of ECL response from this maximal readout was directly proportional to the extent of competitive binding activity in the test sample.

2.4.2.4. Live-Virus Neutralization Assay: NanoLuc Virus Reporter Assay

Three weeks after the second vaccination, live-virus neutralization was assessed by NanoLuc reporter assay. A full-length SARS-CoV-2 virus based on the Seattle, Washington isolate was designed to express luciferase and was recovered via reverse genetics as described previously (Yount et al 2003; Scobey et al 2013; Hou et al 2020). Virus titrations in Vero E6 cells from ATCC that had been in culture at the United States Army Medical Research Institute of Infectious Disease (USAMRIID) were used to obtain a relative light units signal of at least 10× the cell-only control background. Vero E6 USAMRIID cells were plated at 20,000 cells per well in clear bottom black walled 96-well plates. On the next day, neutralizing antibody serum samples were tested at a starting dilution of 1:50 and were serially diluted 2-fold up to 10 dilution spots. Antibody-virus complexes were incubated at 37°C with 5% carbon dioxide for 1 hour. Following incubation, the media was removed, and virus-antibody dilution complexes were added to the cells in duplicate. Virus-only controls and cell-only controls were included in each neutralization assay plate. Following infection, plates were incubated at 37°C with 5% carbon dioxide for 24 hours. After the 24-hour incubation, cells were lysed, and luciferase activity was measured via (b) (4) Luciferase Assay System (b) (4), according to manufacturer specifications. SARS-CoV-2 neutralization titers were defined as the sample dilution at which a 50% reduction in relative light units was observed relative to the average of the virus control wells.

2.4.3. Human Convalescent-Phase Serum

Human convalescent-phase sera (n = 42) were obtained from adults between 18 and 84 years old with mild, moderate, or severe COVID-19 and a history of laboratory-confirmed SARS-CoV-2

infection approximately 1 to 2 months before providing specimens. These specimens were obtained under institutional review board-approved sample collection protocols at the NIH Clinical Center, Bethesda, MD (NCT00067054), Aaron Diamond AIDS Research Center, Columbia University, New York, NY (NCT04342195), and the University of Washington, Seattle, WA (HAARVI study and STUDY00000959). Written informed consent was provided by the participants.

2.4.4. Quantification of SARS-CoV-2 RNA in Bronchoalveolar Lavage Fluid and Nasal Swab Samples

SARS-CoV-2 in the lungs was quantified by plaque assay. Polymerase chain reaction was used to determine viral RNA load in BAL fluid and NS samples; both methods are previously described (Yu et al 2020). To quantify replicated RNA in BAL fluid and NS samples, sgRNA primers were utilized, as described in previously published methods (Wölfel et al 2020).

2.4.5. Intracellular Cytokine Staining

Cryopreserved peripheral blood mononuclear cells were thawed and rested overnight at 37°C in 5% carbon dioxide. Cells were then stimulated with SARS-CoV-2 S protein (S1 and S2, homologous to the vaccine insert) and nucleoprotein overlapping peptide pools (b) (4) at a final concentration of 2 µg/mL/each peptide in the presence of monensin and co-stimulatory antibodies anti-CD28 and anti-CD49d (b) (4) for 6 hours. Negative controls received an equal concentration of dimethyl sulfoxide (instead of peptides) and co-stimulatory antibodies. Intracellular cytokine staining was performed as described (Donaldson et al 2019) except the following monoclonal antibodies were added: PD-1 BUV737 (b) (4) in place of PD-1 BV785, TNF-FITC (b) (4) in place of IL-5 BB515, and CD40L BV785 (b) (4) (b) (4) was used to exclude dead cells. All antibodies were previously titrated to determine the optimal concentration. Samples were acquired on an (b) (4) flow cytometer and analyzed using FlowJo version 9.9.6 (Treestar, Inc.).

2.4.6. Histopathology

Lung tissues from SARS-CoV-2–infected rhesus macaques were processed per a standard protocol. Briefly, the tissues were fixed in 10% neutral buffered formalin, processed with a (b) (4) tissue processor (b) (4), embedded in paraffin, and sectioned at 5 µm for histological analysis. Tissue sections were stained with hematoxylin and eosin for routine histopathology.

2.4.6.1. Immunohistochemistry

Immunohistochemical staining was performed using the formalin-fixed paraffin-embedded tissue sections (5 µm) obtained from lung tissues. Immunohistochemical staining used a rabbit polyclonal SARS-CoV-2 (b) (4) at a dilution of 1:2000. Staining was carried out on the (b) (4) platform according to manufacturer-supplied protocols. Briefly, 5-µm-thick sections were deparaffinized and rehydrated. Heat-induced epitope retrieval was performed using (b) (4), pH 6.0, heated to 100°C for 20 minutes. Specimens were then incubated with hydrogen peroxide to quench endogenous peroxidase activity prior to applying the primary antibody. Detection with 3,3' diaminobenzidine chromogen was completed using a (b) (4), (b) (4). Slides were finally cleared through gradient alcohol and xylene washes prior to mounting and placing cover slips.

All tissue sections were examined by a board-certified veterinary pathologist using an (b) (4) light microscope, and photomicrographs were taken using an (b) (4) camera.

2.4.6.2. In Situ Hybridization

Chromogenic in situ hybridization was performed using RNAScope technology. The target probe detecting positive-sense SARS-CoV-2 (b) (4) was designed by (b) (4). Epitope retrieval was performed on tissues with an application of (b) (4) for 15 minutes at 95°C followed by a 15-minute digestion with (b) (4) (b) (4) at 40°C. Tissues were hybridized with the probes for 2 hours and visualized with the (b) (4) (b) (4) according to the manufacturer's protocol.

2.5. Data Presentation and Analysis

The primary endpoints of the study were reduction of viral load in the BAL fluid of the mRNA-1273 groups compared to the control group. The efficacy endpoint analysis used RT-qPCR and sgRNA RT-qPCR, similar to other nonhuman primate SARS-CoV2 vaccine studies (Munster et al 2020; van Doremalen et al 2020) to quantify viral RNA and replicating virus, respectively. In the BAL fluid and NS specimens, the prespecified primary analysis measured peak viral load over Days 1 through 7 for each animal. Outputs were compared with Mann-Whitney tests and a hierarchical testing procedure.

Statistical analyses were performed in R version 4.0. For all other measures, groups were compared by Kruskal-Wallis test, followed by pairwise Mann-Whitney tests with Holm's adjustment on the set of pairwise tests if the Kruskal-Wallis test indicated significance.

Intracellular cytokine response rates were determined by implementing the Mixture Models for Single-Cell Assays (MIMOSA) ([Finak et al 2013](#)) algorithm.

Graphs were generated using Prism v8 (GraphPad Software). Unless otherwise noted, all graphs are represented as box and whisker plots depicting the median by a horizontal line, the top and bottom of the box in the interquartile range, and the whiskers the range. Intracellular cytokine staining response rates were estimated using the MIMOSA package, which uses a hierarchical beta-binomial model to calculate the probability that each individual is responding to a specific stimulation; values that were judged to be a probable response, subject to a 0.05 false discovery rate constraint, are considered responses for these analyses. Symbols represent individual animals and dashed lines indicate the assay limit of detection.

3. RESULTS

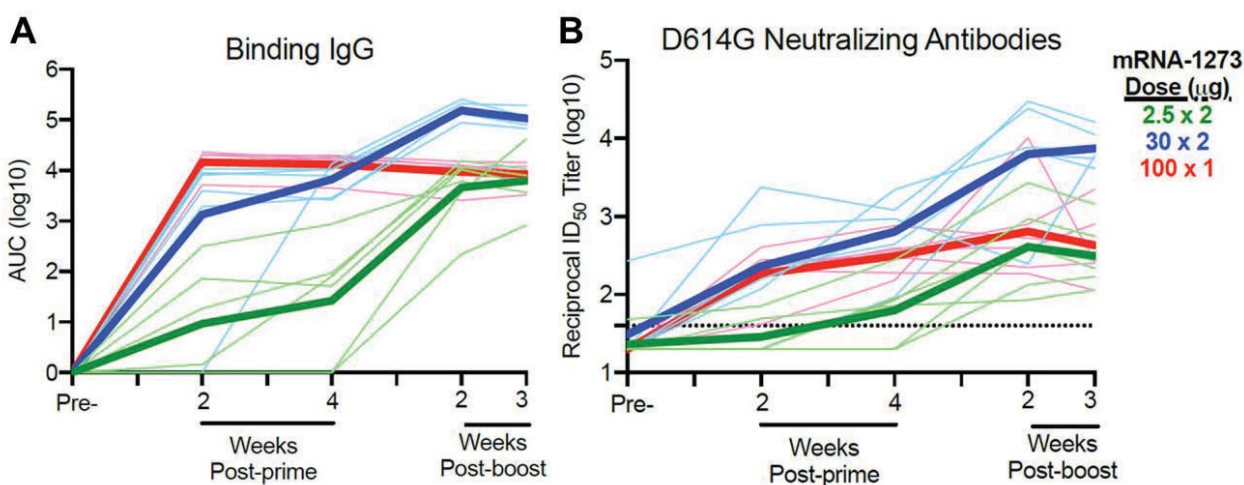
3.1. Binding Antibody Responses and Neutralizing Activity After mRNA-1273 Vaccination

3.1.1. Antibody and Neutralizing Antibody Response

Neutralizing antibodies were measured using a live-virus assay (NanoLuc Virus Reporter Assay, [Section 2.4.2.4](#)) and a pseudovirus assay ([Section 2.4.2.1](#)). These assays are commonly used to assess neutralizing activity in studies for a variety of vaccine candidates.

At all dose levels and dosing schedules (prime/boost and prime only), robust mRNA-1273-induced binding antibody responses were seen in rhesus macaques, and these responses increased in a dose-dependent manner ([Figure 1A](#)). Moreover, post-boost antibody titers were higher than post-prime antibody titers in the 2.5 and 30 μ g groups. A robust neutralizing antibody response against the pseudovirus was observed in rhesus macaques vaccinated with 30 μ g of mRNA-1273 post-boost, and to a lesser degree in the 2.5 μ g prime/boost and 100 μ g prime-only groups ([Figure 1B](#)).

Figure 1: Spike Binding Antibody and Neutralizing Antibody Response



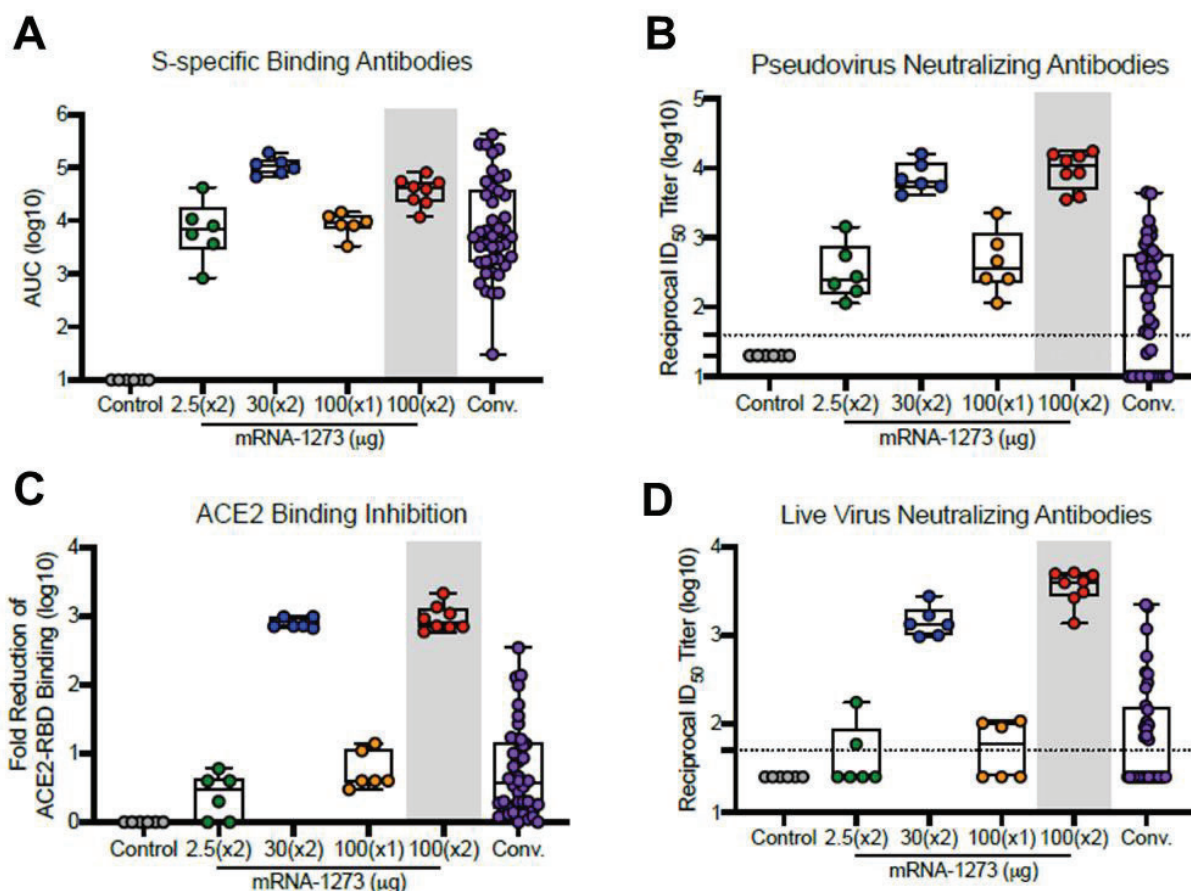
Abbreviations: AUC = area under the curve; D614G = missense mutation in the spike protein of SARS-CoV-2; ID₅₀ = 50% inhibitory dilution; IgG = immunoglobulin G.

Note: Faint lines represent individual animals, and bold lines represent the geometric mean titer for each group.

All mRNA-1273-vaccinated treatment groups had S-specific binding antibody and ACE-2 competing antibody levels that were similar to or higher than those of human convalescent sera ([Figure 2A](#) and [Figure 2C](#), respectively). Neutralizing activity measured against a SARS-CoV-2 pseudovirus assay was observed in all groups but was only evident in the 30 and 100 μ g

prime/boost groups using the live-virus neutralization assay (Figure 2B and Figure 2D, respectively). Animals vaccinated with 30 or 100 μ g of mRNA-1273 on a prime/boost schedule demonstrated the strongest responses compared with human convalescent sera across all assays. The 100 μ g prime/boost group is shaded in Figure 2 because the data were collected in VRC-20-857.1 (Report VRC04).

Figure 2: Antibody and Neutralizing Responses Versus Convalescent Sera



Abbreviations: ACE2 = angiotensin-converting enzyme 2; AUC = area under the curve; Conv. = human convalescent sera; ID₅₀ = reciprocal 50% inhibitory dilution; RBD = receptor binding domain; S = spike.

Note: Noncoding mRNA (NTFIX) formulated into the same lipid nanoparticle dispersion as mRNA-1273 was used as the control. Data contained in the shaded area were collected in VRC-20-857.1 (Report VRC04).

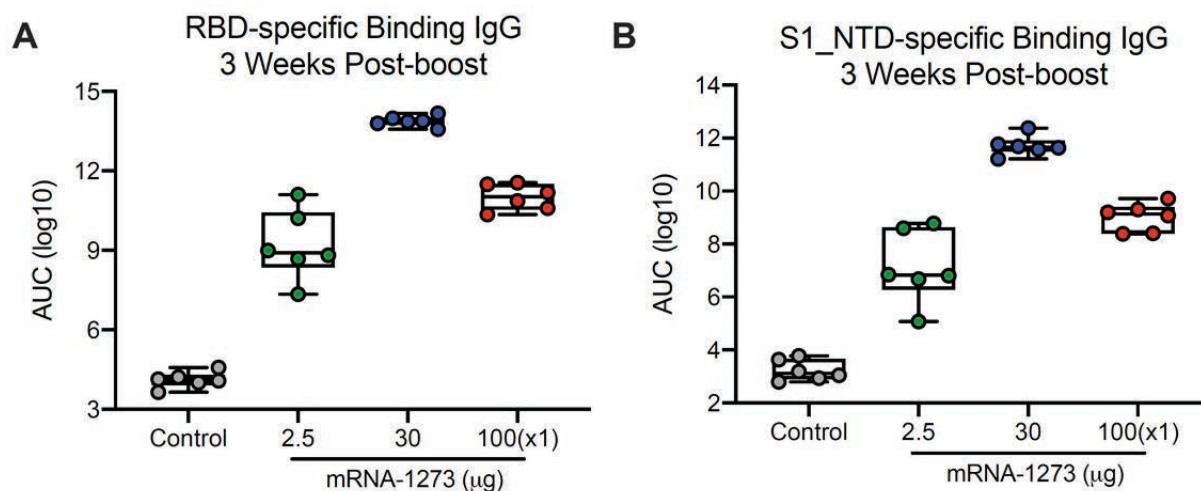
3.1.2. Receptor Binding Domain and N-Terminal Domain Binding Antibody Titers

Fragments containing the functional domains on the S protein (ie, RBD and NTD) may also be used as antigens to create antibodies to block virus binding or fusion, creating a polyclonal antibody response. Furthermore, the elicitation of antibodies to epitopes (eg, RBD and NTD), in addition to the whole S protein, could mitigate the possibility of immune escape by antigenic

drift (Wang et al 2018). Therefore, analysis of antibody response was extended to these specific domains of the S protein.

Compared with the 2.5 μ g mRNA-1273 prime/boost group, higher anti-RBD and anti-S1_NTD responses were observed in the 30 μ g prime/boost and 100 μ g prime-only groups (Figure 3). These trends were similar to what was observed for the S-specific and neutralizing antibody responses.

Figure 3: Receptor Binding Domain and N-Terminal Domain of S1 Binding Antibody Titers

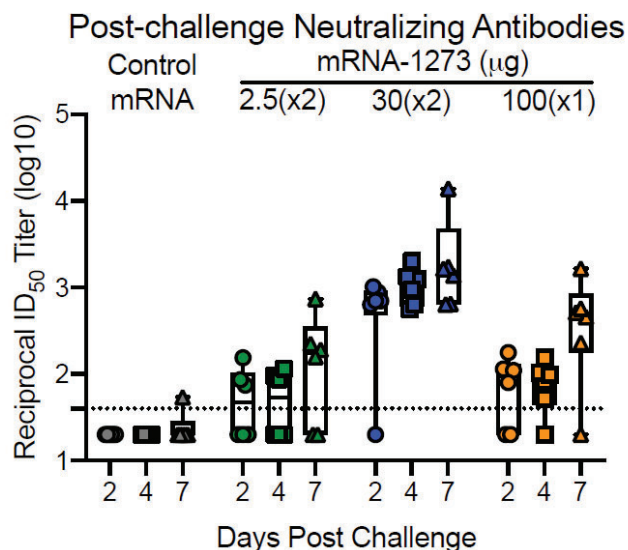


Abbreviations: AUC = area under the curve; IgG = immunoglobulin G; S1_NTD = N-terminal domain of S1; RBD = receptor binding domain.

Note: Noncoding mRNA (NTFIX) formulated into the same lipid nanoparticle dispersion as mRNA-1273 was used as the control.

3.1.3. Post-challenge Neutralization Titers

Neutralizing activity was also assessed post-challenge. Neutralizing antibodies were substantially higher in the 30 μ g mRNA-1273 prime/boost group at 2 days, 4 days, and 7 days post-challenge than in the control group and the 2.5 μ g prime/boost group and were somewhat higher than the 100 μ g prime-only group at 7 days post-challenge (Figure 4).

Figure 4: Post-challenge Neutralization Titers

Abbreviations: ID₅₀ = 50% inhibitory dilution.

Note: Noncoding mRNA (NTFIX) formulated into the same lipid nanoparticle dispersion as mRNA-1273 was used as the control.

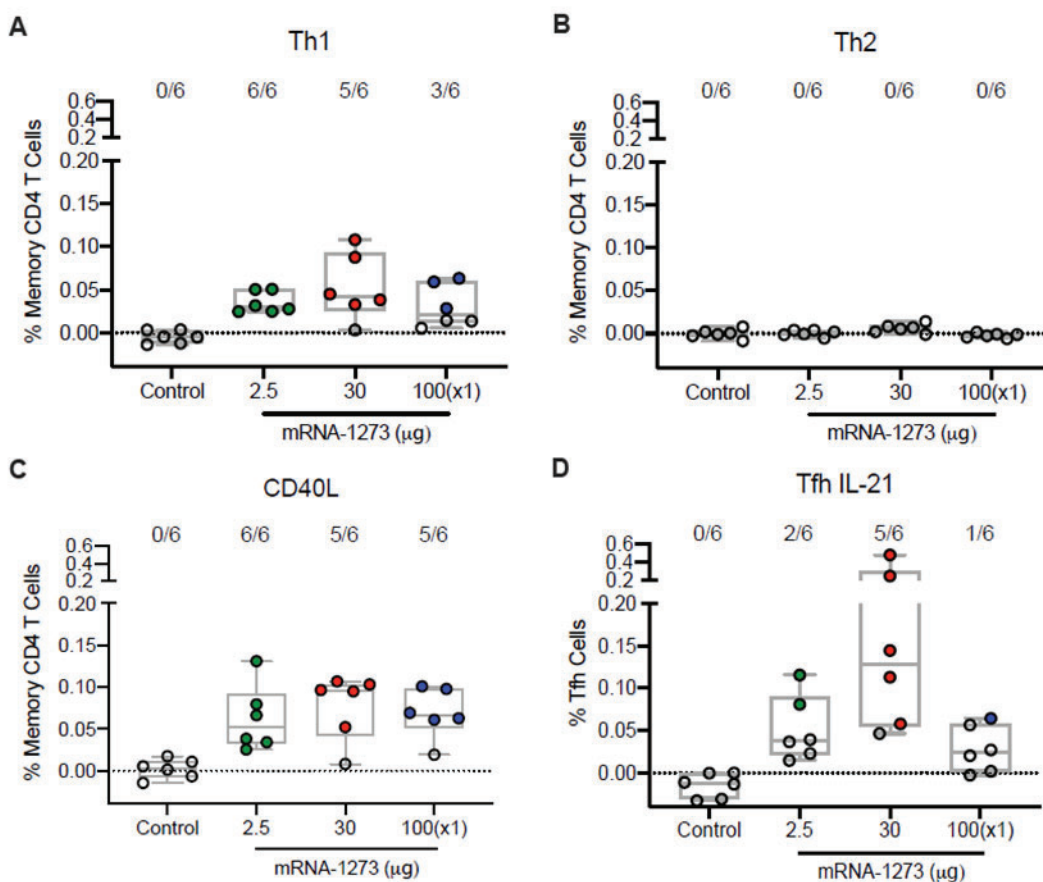
3.2. T-Cell Responses Following mRNA-1273 Vaccination

Studies have shown that SARS-CoV-2-specific T-cell immunity may have a role in pathogenesis or protection against SARS-CoV-2 and may influence the humoral immune response (Meckiff et al 2020; Neidleman et al 2020; Weiskopf et al 2020). Activated CD4⁺ T cells can be segregated into Th1- and Th2-directed responses based on the production of specific cytokines and are critical for B cell activation and antibody production. Importantly, the induction of CD4⁺ Th2 T-cell responses (production of IL-4, IL-5, or IL-13) has been linked to vaccine-associated enhanced respiratory disease, as seen in some human respiratory virus infections (eg, respiratory syncytial virus [Kim et al 1969], measles [Polack 2007], and animal models of SARS-CoV [Bolles et al 2011; Corbett et al 2020; Deming et al 2007]); however, vaccine-associated enhanced respiratory disease has not been observed with exclusive CD4⁺ Th1 T-cell responses (production of interferon- γ , IL-2, and tumor necrosis factor- α) (Grifoni et al 2020; Peng et al 2020; Sekine et al 2020; Weiskopf et al 2020).

Intracellular cytokine staining was performed on peripheral blood mononuclear cells collected 6 weeks post-prime (2 weeks before viral challenge) to assess T-cell responses to the SARS-CoV-2 S1 peptide pool. A 19-color multiparameter flow cytometry was used to assess the functional heterogeneity of S-specific CD4⁺ and CD8⁺ T-cell cytokine responses. A dose-dependent increase in Th1 responses was observed in 5 of 6 animals in the

30 µg mRNA-1273 prime/boost group, compared with control and the 2.5 µg mRNA-1273 prime/boost group. Overall, Th1-directed CD4⁺ T-cell response levels were highest in the 30 µg prime/boost group but were measured in all of the mRNA-1273 groups (Figure 5A) and Th2 responses were low to undetectable (Figure 5B). CD40L is a cell-surface marker expressed after CD4⁺ T-cell activation and mediates B-cell activation for efficient isotype switching. Results showed that 6 of 6 animals in the 2.5 µg prime/boost group, 5 of 6 animals in the 30 µg prime/boost group, and 5 of 6 animals in the 100 µg prime-only group had S-specific CD40L + CD4⁺ T-cell responses (Figure 5C). Nucleoside-modified RNA vaccines that encode various viral antigens have been shown to induce robust antibody responses in rhesus macaques and are associated with increased CD4⁺ Tfh cells (Rogers et al 2020). Therefore, IL-21 (the canonical cytokine produced by Tfh cells) was measured. Two of 6 animals in the 2.5 µg group, 5 of 6 animals in the 30 µg group, and 1 of 6 animals in the 100 µg prime-only group had IL-21 responses. The magnitude of responses was greater in the 30 µg group than in the control, 2.5 µg prime/boost, or 100 µg prime-only group, and the response in the 2.5 µg prime/boost and 100 µg prime-only group was greater than that in the control group (Figure 5D).

These data demonstrate that mRNA-1273 induced Th1-directed CD4⁺ T-cell and IL-21-producing Tfh cell responses with no evidence of Th2-directed CD4⁺ T-cell responses.

Figure 5: CD4+ T Cells Following mRNA-1273 Vaccination in Rhesus Macaques

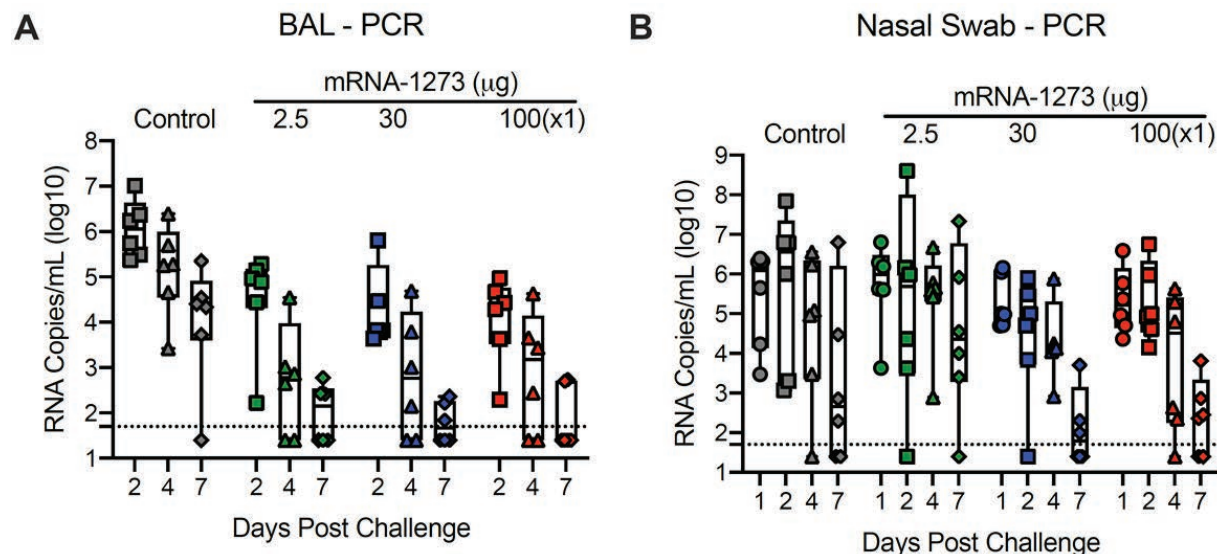
Abbreviations: Tfh = T-follicular helper; Th1 = T-helper 1; Th2 = T-helper 2.

Note: Noncoding mRNA (NTFIX) formulated into the same lipid nanoparticle dispersion as mRNA-1273 was used as the control. Open symbols represent animals with a probable nonresponse, and solid symbols represent animals with a probable response.

3.3. Protective Efficacy Against Upper and Lower SARS-CoV-2 Infection

In order to evaluate the protective efficacy of mRNA-1273 vaccination, all animals were challenged IT/IN with SARS-CoV-2 at 4 weeks post-boost (8 weeks post-prime for the prime-only group). Quantification of SARS-CoV-2 was performed using PCR to determine viral RNA load in BAL fluid and NS samples. To differentiate between input challenge virus and newly replicated virus, primers specific for sgRNA were used in the RT-qPCR assay.

All mRNA-1273 groups demonstrated lower viral loads in BAL fluid than the control group (Figure 6A). Differences in viral load in NS samples between control and mRNA-1273 groups were less obvious, although animals in the 30 µg prime/boost group trended towards having lower viral titers than those in other groups (Figure 6B).

Figure 6: Viral Load as Measured by PCR

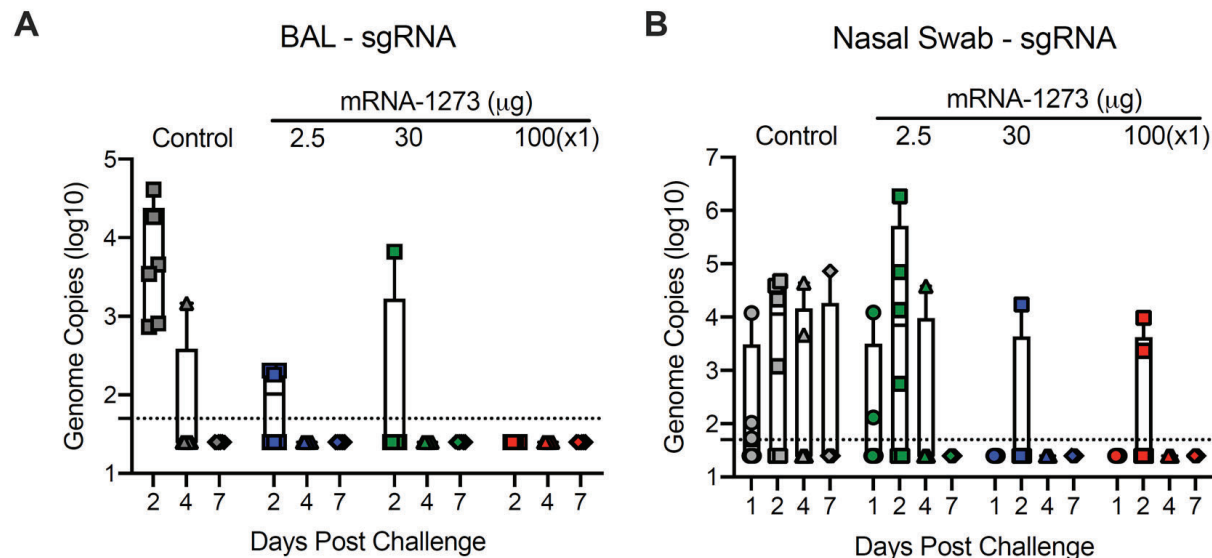
Abbreviations: BAL = bronchoalveolar lavage; PCR = polymerase chain reaction.

Note: Noncoding mRNA (NTFIX) formulated into the same lipid nanoparticle dispersion as mRNA-1273 was used as the control.

In the 30 μg prime/boost group, mRNA-1273 was protective in the lungs and nose 2 days post-challenge as measured by sgRNA (Figure 7). A prime-only dose of 100 μg of mRNA-1273 was also protective in rhesus macaque.

In the BAL fluid, no animals in the 100 μg prime-only group and only 1 of 6 animals in the 30 μg prime/boost group had detectable sgRNA after challenge, compared with 3 of 6 animals in the 2.5 μg prime/boost group and all 6 animals in the control group (Figure 7A). The breakthrough observed in the 2.5 μg prime/boost group will aid in establishing a surrogate of protection.

In NS samples, no animals in the 30 μg prime/boost or the 100 μg prime-only group had detectable sgRNA at 1 day and 4 days post-challenge (Figure 7B). Only 1 of 6 animals had detectable levels of sgRNA at 2 days post-challenge in the 30 μg prime/boost group, and 2 of 6 animals had detectable levels of sgRNA at 2 days post-challenge in the 100 μg prime-only group.

Figure 7: Bronchoalveolar Lavage and Nasal Swab Fluid sgRNA Levels

Abbreviations: BAL = bronchoalveolar lavage; sgRNA = subgenomic RNA.

Note: Noncoding mRNA (NTFIX) formulated into the same lipid nanoparticle dispersion as mRNA-1273 was used as the control.

3.4. Histopathology and Viral Load in Lung Tissue Post-challenge

At 7, 8, and 9 days after challenge, lungs were harvested from 2 animals per group for histopathologic analysis and assessment for evidence of viral infection.

On both Day 7 and Day 8, animals in the control group showed multiple to numerous foci of SARS-CoV-2 positive cells, and inflammation was moderate to severe with a greater number of affected lung sections (Table 3). Of animals vaccinated with mRNA-1273, in the 2.5 µg prime/boost and 100 µg prime-only groups, viral antigen were present but sparse and positive areas were limited to mostly single cells, single foci, or small clusters of cells. Inflammation in these animals ranged from absent to occasional, and there was no correlation between the presence or location of inflammation and viral detection. In the 30 µg prime/boost group, no viral antigen was detected at either timepoint and inflammation ranged from absent to occasional. At Day 9, except for one animal in the control group, viral antigen was not detected.

Overall, no evidence of vaccine-associated immunopathology was found in histopathological analysis of lung tissue and no viral antigen was detected in the 30 µg prime/boost group post-challenge.

Table 3: Assessment of Lung Inflammation, Viral RNA, and Viral Antigen Following SARS-CoV-2 Challenge of mRNA-1273-Immunized Rhesus Macaques

Day Post-Challenge	mRNA Dose (µg)	Inflammation (H&E)	IHC ^a	Antigen Distribution
7	Control mRNA	Moderate-severe	+ ; + ; -	Multiple positive foci
7	Control mRNA	Mild-occasional	+/- ; +/- ; +	Multiple positive foci
7	2.5	Mild-occasional	+/- ; - ; -	1 to 2 small clusters of positive cells
7	2.5	Mild-occasional	- ; - ; +/-	Single positive cell
7	30	Absent-minimal	- ; - ; -	NA
7	30	Mild-occasional	- ; - ; -	NA
7	100 (prime only)	Mild-occasional	- ; - ; -	NA
7	100 (prime only)	Mild-occasional	+/- ; - ; -	Single positive cell
8	Control mRNA	Moderate-severe	+/- ; + ; +	Multiple positive foci
8	Control mRNA	Moderate-severe	- ; + ; ++	Numerous positive foci
8	2.5	Mild-occasional	- ; - ; -	NA
8	2.5	Absent-minimal	+/- ; - ; -	1 to 2 small clusters of positive cells
8	30	Mild-occasional	- ; - ; -	NA
8	30	Mild-occasional	- ; - ; -	NA
8	100 (prime only)	Absent-minimal	- ; - ; -	NA
8	100 (prime only)	Absent-minimal	- ; - ; +/-	Single positive cell
9	Control mRNA	NR	- ; - ; -	NA
9	Control mRNA	NR	+/- ; +/- ; -	NR
9	2.5	NR	- ; - ; -	NA
9	2.5	NR	- ; - ; -	NA
9	30	NR	- ; - ; -	NA
9	30	NR	- ; - ; -	NA
9	100 (prime only)	NR	- ; - ; -	NA
9	100 (prime only)	NR	- ; - ; -	NA

Abbreviations: H&E = hematoxylin & eosin; IHC = immunohistochemistry; mRNA = messenger RNA; NA = not applicable; NR = not reported; SARS-CoV-2 = 2019 novel coronavirus.

^a IHC antigen detection scoring: Left caudal ; right middle ; right caudal. - = no detection of viral antigen; +/- = rare/minimal; + = moderate/moderate; ++ = multiple/numerous.

4. CONCLUSIONS

Results from this study showed that mRNA-1273 elicits robust binding and neutralizing antibody responses in rhesus macaques, especially in the 30 µg prime/boost group. Similar antibody responses were observed in the RBD and NTD domains of the S1 subunit. Vaccination with mRNA-1273 induced Th1-directed T-cell and IL-21-producing Tfh cell responses, a profile that is not predicted to promote ERD. mRNA-1273 was protective in the lungs and nose of animals in the 30 µg prime/boost and 100 µg prime-only groups, as measured by sgRNA. No evidence of vaccine-associated immunopathology was found in histopathological analysis of lung tissue of animals vaccinated with mRNA-1273, and no viral antigen was detected in the 30 µg prime/boost group after challenge.

Overall, immunization of rhesus macaques with mRNA-1273 induced robust SARS-CoV-2 binding antibodies and neutralizing activity and protection in the upper and lower airways from WT SARS-CoV-2 challenge, with no pathologic changes observed in the lungs or signs of disease enhancement.

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6. REPORT APPROVAL

Study Director:

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Date: 2020.11.05 11:59:36 -05'00'

Date and Signature

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Date and Signature

(b) (6)

Viral Pathogenesis Laboratory, VRC, NIH
Building 40 Room 2608
Bethesda, MD 20892

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Viral Pathogenesis Laboratory, VRC, NIH
Building 40 Room 2608
Bethesda, MD 20892

7. APPENDICES

Appendix 1 Summary of Analysis: Lot AMPDP-200005 Release



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	(b) (4)
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	(b) (4)
RNA Purity	DPTM-0046	(b) (4)	(b) (4)
% Encapsulation	DPTM-0063	(b) (4)	(b) (4)
Filtered Mean Particle Size	DPTM-0012	(b) (4)	(b) (4)
Filtered Polydispersity Index		(b) (4)	(b) (4)



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Test	Method	Target Attributes	Results
Lipid Identification:		(b) (4)	
SM-102			
Cholesterol			
DSPC	DPTM-0019		
PEG2000-DMG			
Lipid Content (mg/mL)			
SM-102		(b) (4)	(b) (4)
Cholesterol			
DSPC	DPTM-0019		
PEG2000-DMG			
Lipid Impurities (% Area)			
Individual Impurities	DPTM-0019	(b) (4)	
Total Impurities			
pH	USP <791> DPTM-0028		
Osmolality	USP <785> DPTM-0029		
Bacterial Endotoxin	USP <85>		
Bioburden	USP<61>		
Particulate Matter	USP <788> Method 2		



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval